

1. Motivation & Introduction

Remark

- Gaussian Process Regression (GPR) is a nonlinear regression method.
- Gaussian Processes (GP) are a powerful modelling assumption with a lot useful properties (as always with Gaussians).
- The key advantage is the uncertainty quantification for predictions.

The main idea is:

① assume our data come from a GP, which is defined by

 Ⓐ a mean function m , which encodes the general behaviour of the process at different locations,

 Ⓑ a covariance function (kernel) k , which determines the relationship between different locations of the process,

② fit mean and kernel to data,

③ obtain posterior predictive distribution for all other location,

[④ give an acquisition function, sample next point.] ^{Bayesian optimisation}

GPR is used in

- machine learning
 - for hyperparameter search / active learning,
 - as a world model in reinforcement learning,
- geostatistics,
- time series modelling.

Agenda

- Gaussian Distribution $\rightarrow$ Gauss Process
- Kernels
- Model Fitting
- Break
- Posterior Distribution
- Bayesian Optimisation

Literature:

Gaussian Processes for Machine Learning, Rasmussen & Williams, 2006

2. Gaussian Distributions and Processes

Def (Multivariate Gaussian)

Given $\mu \in \mathbb{R}^n$ and an invertible, symmetric and positive semi-definite $\Sigma \in \mathbb{R}^{n \times n}$, the multivariate normal distribution $N(\mu, \Sigma)$ on $\mathbb{R}^n$ has the density

$$f(x) = \frac{1}{\sqrt{(2\pi)^n \det(\Sigma)}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

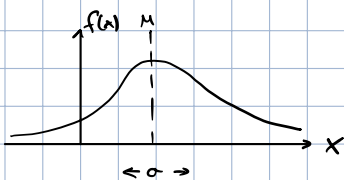
for all $x \in \mathbb{R}^n$.

Remark (Properties)

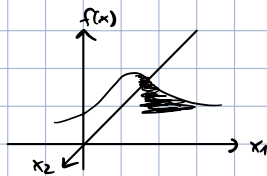
For $X \sim N(\mu, \Sigma)$, it holds that

- $E[X_i] = \mu_i$, $\text{Cov}(X_i, X_j) = \Sigma_{ij}$ for all $i, j \in \{1, \dots, n\}$,

- $n=1$:



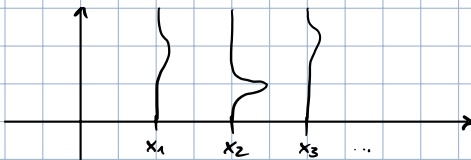
- $n=2$:



or



- $n > 2$ and $n \rightarrow \infty$



Def (Stochastic processes on $\mathbb{R}$)

Let $I \subseteq \mathbb{R}$ be an index. A family of random of real-valued random variables $X = (X_t)_{t \in I}$ is a stochastic process.

Def (Gaussian process)

A stochastic process $(X(t))_{t \in \mathbb{R}}$ is a Gaussian process (GP) if $(X(t_1), \dots, X(t_n))$ is normally distributed for all finite sets $\{t_1, \dots, t_n\} \subseteq \mathbb{R}$.

Remark

A GP is uniquely defined via a mean function $m: \mathbb{R} \rightarrow \mathbb{R}$ and a kernel function $k: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, s.t., $(X(t_1), \dots, X(t_n)) \sim N(\mu, \Sigma)$, where

$$E[X(t_i)] = \mu_i = m(t_i) \text{ and } \text{Cov}(X(t_i), X(t_j)) = \Sigma_{ij} = k(t_i, t_j) \text{ for all } i, j \in \{1, \dots, n\}. \text{ We write } X \sim \text{GP}(m, k).$$

As we want to approximate $f: \mathbb{R}^d \rightarrow \mathbb{R}$, we switch the following

- $t \in \mathbb{R} \rightarrow x \in \mathbb{R}^d$, $X(t) \rightarrow f(x)$, $m: \mathbb{R}^d \rightarrow \mathbb{R}$, $k: \mathbb{R}^{d \times d} \rightarrow \mathbb{R}$,

- $f(x) \sim \text{GP}(m, k) \Leftrightarrow (f(x_1), \dots, f(x_n)) \sim N(\mu, \Sigma)$, $\mu = (m(x_1), \dots, m(x_n))^T$, $\Sigma = (k(x_i, x_j))_{i,j=1}^n$.

3 Kernels

Def (kernel function)

A symmetric function $k: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a kernel function (or kernel) if it is positive semi-definite, i.e.,

$$\sum_{i=1}^d \sum_{j=1}^d c_i c_j k(x_i, x_j) \geq 0$$

for all $x_1, x_2 \in \mathbb{R}^d$, $c_1, \dots, c_n \in \mathbb{R}$. (Positive definite additionally imposes equality if and only if $c_i = 0$ for all $i \in \{1, \dots, d\}$.)

Example	stationary ($K(x-y)$)	isotropic ($K(\ x-y\)$)	(semi-) positive definite	other properties
linear kernel $k(x, y) = \sum_{i=1}^d w_i x_i y_i$	no	no	spd	
polynomial kernel $k(x, y) = (x^T y + r)^m, \quad m \geq 1$	no	no	spd	
Squared exponential (RBF) $k(x, y) = \sigma_f^2 \exp\left(-\frac{1}{2} \frac{\ x-y\ ^2}{\ell^2}\right)$	yes	yes	pd	smooth
automatic relevance determination $k(x, y) = \sigma_f^2 \exp\left(-\frac{1}{2} (x-y)^T \Lambda^{-1} (x-y)\right),$ $\Lambda = \text{diag}(\ell_1^2, \dots, \ell_d^2),$	yes	no	pd	
Matérn $k(x, y) = \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}}{\rho} \ x-y\ ^2\right)^\nu \mathcal{B}_\nu \left(\frac{\sqrt{2\nu}}{\rho} \ x-y\ ^2\right),$ $\rho, \nu \geq 0$, Gamma function Γ, Bessel function $\mathcal{B}_\nu$	yes	yes	pd for all $\nu > 0$	L^ν -differentiable

Remark

- Nonnegative scaling, the sum, the product, ... of kernel functions are again kernels.
- For $\nu = \frac{1}{2}$ Matérn is exponential kernel (not given) and for $\nu \rightarrow \infty$ Matérn kernel is squared exponential kernel.
- (informal) If a kernel is 2ν -times differentiable (+Lipschitz), the sample paths of the GP are (a.s.) ν -times differentiable. (Da Costa, 2023)
- There exist a strong connection to kernelised Bayesian linear regression.

4 Model Fitting

How do we find the best parameters η for a chosen kernel k , e.g., σ_f^2, ℓ for RBF kernel?

Assumption: we have the following

- training data $\mathcal{D} = \{(x_i, f(x_i))\}_{i=1}^n$
- a choice of a kernel k with hyperparameters η .

The log-likelihood for $\bar{f} \sim \mathcal{N}(\mu, K)$ is given by

$$-\frac{1}{2} \left((\bar{f} - \mu)^T K^{-1} (\bar{f} - \mu) + \log(\det(K)) + n \log(2\pi) \right),$$

where $\bar{f} = (f(x_1), \dots, f(x_n))^T \in \mathbb{R}^n$, $K = (k(x_i, x_j))_{i,j=1}^n \in \mathbb{R}^{n \times n}$.

The mean function is often assumed to be constant $m(x) = m^*$.

This leads to

$$m^* = \left(\frac{\mathbf{1}^T K^{-1} \bar{f}}{\mathbf{1}^T K^{-1} \mathbf{1}} \right) \mathbf{1},$$

where $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^n$.

For the kernel function we maximise with the gradient (assuming $\mu=0$)

$$\frac{\partial}{\partial \eta} \log(\mathbb{P}(\bar{f} | \eta)) = \frac{1}{2} (\bar{f}^T K(\eta)^{-1} \frac{\partial K(\eta)}{\partial \eta} K(\eta)^{-1} \bar{f}) - \frac{1}{2} \text{tr} \left(K(\eta)^{-1} \frac{\partial K(\eta)}{\partial \eta} \right),$$

where $K(\eta)$ is the covariance matrix dependent on η .

Remark

Other methods include

- maximum a posteriori,
- full Bayesian treatment.

The simulation of a GP on a grid $\{x_i\}_{i=1}^n$ works as follows:

- ① compute K and its Cholesky decomposition $K = LL^T$
- ② draw n i.i.d. $z \sim \mathcal{N}(0, 1)$ and set $\bar{f} = \mu + L\bar{z}$.

Programming examples: kernels, GP regression

Break

4 Posterior Distribution

Given data $\{(x_i, f(x_i))\}_{i=1}^n$ and a new location x_{n+1} , how can we sample $f(x_{n+1})$ while preserving the dependence structure?

Theorem

If

$$\begin{pmatrix} \bar{f} \\ f_{n+1} \end{pmatrix} \sim \mathcal{N} \left(0, \begin{pmatrix} K & k(x_{n+1}) \\ k^T(x_{n+1}) & k(x_{n+1}, x_{n+1}) \end{pmatrix} \right),$$

where $\bar{f} = (f(x_1), \dots, f(x_n))^T \in \mathbb{R}^n$, $K = (k(x_i, x_j))_{i,j=1}^n \in \mathbb{R}^{n \times n}$ and $k(x_{n+1}) = (k(x_{n+1}, x_1), \dots, k(x_{n+1}, x_n))^T \in \mathbb{R}^n$,

then

$$f_{n+1} \mid \{(x_i, f(x_i))\}_{i=1}^n, x_{n+1} \sim \mathcal{N}(\mu_n(x_{n+1}), \sigma_n^2(x_{n+1}))$$

where $\mu_n(x_{n+1}) = k^T(x_{n+1}) K^{-1} \bar{f}$, $\sigma_n^2(x_{n+1}) = k(x_{n+1}, x_{n+1}) - k^T(x_{n+1}) K^{-1} k(x_{n+1})$.

Remark

- The theorem can be extended to multiple values $x_{n+1}, \dots, x_{n+m}$.
- The joint has zero mean, the conditional not.
- The variance does not depend on $\bar{f}$.
- The conditional is exact at the given data points, because

- the mean is

$$\mu_n(x_i) = k^T(x_i) K^{-1} \bar{f} = k_i \cdot K^{-1} \bar{f} = e_i^T \bar{f} = f_i \text{ for all } i \in \{1, \dots, n\},$$

- the variance is

$$\sigma_n^2(x_i) = k(x_i, x_i) - k^T(x_i) K^{-1} k(x_i) = k(x_i, x_i) - k(x_i, x_i) = 0.$$

- If we assume observation noise, i.e., $y_i = f(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_{\text{noise}}^2)$, for all $i \in \{1, \dots, n\}$,

we get analogous results:

- hyperparameter: use extended log-likelihood (assuming $\mu=0$)

$$-\frac{1}{2} \bar{y}^T (K + \sigma_{\text{noise}}^2 \mathbb{1}) \bar{y} - \frac{1}{2} \log(\det(K + \sigma_{\text{noise}}^2 \mathbb{1})) - \frac{n}{2} \log(2\pi)$$

and optimise,

- predictive posterior:

$$\mu_n(x_{n+1}) = k^T(x_{n+1}) (K + \sigma_{\text{noise}}^2 \mathbb{1})^{-1} \bar{y},$$

$$\sigma_n(x_{n+1}) = k(x_{n+1}, x_{n+1}) - k^T(x_{n+1}) (K + \sigma_{\text{noise}}^2 \mathbb{1})^{-1} k + \sigma_{\text{noise}}^2.$$

Proof of Theorem (Bishop, 2006)

Assume

$$\begin{pmatrix} x_A \\ x_B \end{pmatrix} \sim \mathcal{N} \left(\underbrace{\begin{pmatrix} \mu_A \\ \mu_B \end{pmatrix}}_{\mu}, \underbrace{\begin{pmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{pmatrix}}_{\Sigma} \right),$$

where $\mu_A \in \mathbb{R}^n$, $\mu_B \in \mathbb{R}^m$, $\Sigma_{AA} \in \mathbb{R}^{n \times n}$, $\Sigma_{AB} = \Sigma_{BA}^T \in \mathbb{R}^{n \times m}$, $\Sigma_{BB} \in \mathbb{R}^{m \times m}$,

then

$$x_B | x_A \sim \mathcal{N}(\mu_B + \Sigma_{BA} \Sigma_{AA}^{-1} (x_A - \mu_A), \Sigma_{BB} - \Sigma_{BA} \Sigma_{AA}^{-1} \Sigma_{AB}).$$

① Conditional density

$$\begin{aligned} g(x_B | x_A) &= \frac{g(x_A, x_B | \mu, \Sigma)}{\int_{\mathbb{R}^m} g(x_A, x_B | \mu, \Sigma) dx_B} \\ &= \underbrace{\frac{1}{Z}}_{\text{norm. const.}} \exp \left(-\frac{1}{2} \begin{pmatrix} x_A - \mu_A \\ x_B - \mu_B \end{pmatrix}^T \underbrace{\begin{pmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{pmatrix}^{-1}}_{=: \begin{pmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{pmatrix}} \begin{pmatrix} x_A - \mu_A \\ x_B - \mu_B \end{pmatrix} \right) \quad \text{! } V_{AA} = \Sigma_{AA}^{-1} \end{aligned}$$

② Matrix multiplication and completing the square

$$= \frac{1}{Z} \exp \left(-\frac{1}{2} \left((x_A - \mu_A)^T V_{AA} (x_A - \mu_A) + (x_A - \mu_A)^T \Sigma_{AB} (x_B - \mu_B) + (x_B - \mu_B)^T V_{BA} (x_A - \mu_A) + (x_B - \mu_B)^T \Sigma_{BB} (x_B - \mu_B) \right) \right)$$

$$\left[\text{Completing the square: } \underbrace{\frac{1}{2} z^T A z}_{x_B - \mu_B} + \underbrace{b^T z}_{V_{BA} (x_A - \mu_A)} + c = \frac{1}{2} (z + A^{-1} b)^T A (z + A^{-1} b) + c - \frac{1}{2} b^T A^{-1} b \right]$$

$$= \frac{1}{Z} \exp \left(-\frac{1}{2} \left((x_B - \mu_B + V_{BB}^{-1} V_{BA} (x_A - \mu_A))^T V_{BB} (x_B - \mu_B + V_{BB}^{-1} V_{BA} (x_A - \mu_A)) + (x_A - \mu_A)^T V_{AA} (x_A - \mu_A) - (x_A - \mu_A)^T V_{AB} V_{BB}^{-1} V_{BA} (x_A - \mu_A) \right) \right)$$

$$= \underbrace{\frac{1}{Z}}_{\text{incl. term ind. of } x_B} \exp \left(-\frac{1}{2} (x_B - \mu_B + V_{BB}^{-1} V_{BA} (x_A - \mu_A))^T V_{BB} (x_B - \mu_B + V_{BB}^{-1} V_{BA} (x_B - \mu_B)) \right)$$

Schur complement

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \Rightarrow M^{-1} = \begin{pmatrix} A^{-1} + A^{-1} B (D - C A^{-1} B)^{-1} C A^{-1} & -A^{-1} B (D - C A^{-1} B)^{-1} \\ -(D - C A^{-1} B)^{-1} C A^{-1} & (D - C A^{-1} B)^{-1} \end{pmatrix}$$

This leads to

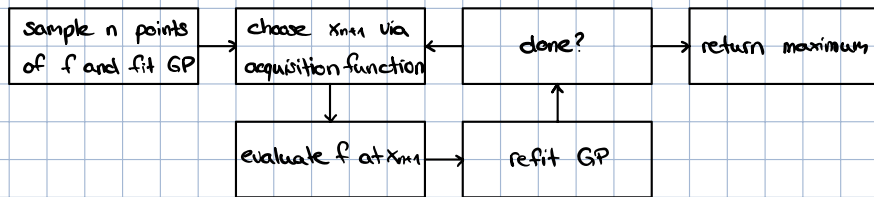
$$\Sigma_{B|A} = V_{BB}^{-1} = \Sigma_{BB} - \Sigma_{BA} \Sigma_{AA}^{-1} \Sigma_{AB}$$

$$\mu_{B|A} = \mu_B - V_{BB}^{-1} V_{BA} (x_A - \mu_A) = \mu_B + \Sigma_{BA} \Sigma_{AA}^{-1} (x_A - \mu_A).$$

5 Bayesian Optimisation

The goal of Bayesian optimisation (Bo) is to maximise a function f , which is expensive to evaluate.

The pipeline is



Def (acquisition function)

A function $a: \mathbb{R}^d \rightarrow \mathbb{R}$ is called acquisition function. The maximiser of a is the next sampling notation.

Examples

- upper confidence bound (UCB)

$$\text{UCB}(x) = \mu(x) + k \sigma(x)$$

- expected improvement (EI)

$$\begin{aligned} \text{EI}(x) &= \operatorname{argmax}_x \mathbb{E} \left[\max \left(0, f_{n+1}(x) - \max_{i=1, \dots, n} f(x_i) \right) \right] \\ &= \left(\mu(x) - \max_{i=1, \dots, n} f(x_i) \right) \Phi \left(\frac{\mu(x) - \max_{i=1, \dots, n} f(x_i)}{\sigma(x)} \right) + \sigma(x) \phi \left(\frac{\mu(x) - \max_{i=1, \dots, n} f(x_i)}{\sigma(x)} \right), \end{aligned}$$

where Φ and ϕ are the standard normal CDF and PDF, respectively.